Electrochemical equilibrium

Week 6 · Onsite · 180 minutes

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Start with the chemical potential

Follow the same reaction throughout the derivation.

$$\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$$

The chemical potential of each species depends on its activity.

$$\mu_i = \mu_i^\circ + RT \ln a_i$$

μ_i° refers to the chosen standard state. The activity a_i is dimensionless. For a gas, use $a_i = f_i / p^\circ$.

Combine the potentials along the reaction

At fixed temperature and pressure, a small forward reaction consumes one unit of A_2 and forms two units of A.

$$\mathrm{d}n_{\mathrm{A}} = 2\,\mathrm{d}\xi, \qquad \mathrm{d}n_{\mathrm{A_2}} = -\mathrm{d}\xi$$

Substitute these amounts into $\mathrm{d}G = \sum_i \mu_i\,\mathrm{d}n_i$.

$$\mathrm{d}G = (2\mu_{\mathrm{A}} - \mu_{\mathrm{A_2}})\,\mathrm{d}\xi$$

The reaction Gibbs energy is the slope along the reaction extent ξ.

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi}\right)_{T,P} = 2\mu_{\mathrm{A}} - \mu_{\mathrm{A_2}}$$

The activity ratio is Q

Insert $\mu_i = \mu_i^\circ + RT\ln a_i$ into the reaction Gibbs energy.

$$\Delta_r G = (2\mu_{\mathrm{A}}^\circ - \mu_{\mathrm{A_2}}^\circ) + RT(2\ln a_{\mathrm{A}} - \ln a_{\mathrm{A_2}})$$

Define $\Delta_r G^\circ = 2\mu_{\mathrm{A}}^\circ - \mu_{\mathrm{A_2}}^\circ$ and combine the logarithms.

$$\Delta_r G = \Delta_r G^\circ + RT\ln\left(\frac{a_{\mathrm{A}}^2}{a_{\mathrm{A_2}}}\right)$$

The ratio calculated from the mixture's **current activities** is the reaction quotient.

$$Q = \frac{a_{\mathrm{A}}^2}{a_{\mathrm{A_2}}}, \qquad \Delta_r G = \Delta_r G^\circ + RT\ln Q$$

At equilibrium, Q equals K

At reaction equilibrium, $\Delta_r G = 0$. Call the equilibrium value of the activity ratio K.

$$0 = \Delta_r G^\circ + RT \ln K, \qquad K = \exp\left(-\frac{\Delta_r G^\circ}{RT}\right)$$

For the written reaction and chosen standard states, the temperature determines K. Combine this relation with the expression for the current mixture.

$$\boxed{\Delta_r G = RT \ln\left(\frac{Q}{K}\right)}$$

Compare Q with K to predict the reaction direction. Which sign does $\Delta_r G$ have when $Q < K$?

Q and K for the same reaction

For $\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$, use the dimensionless activity ratio $a_{\mathrm{A}}^2 / a_{\mathrm{A_2}}$.

Reaction quotient, Q. Calculate the ratio using the activities in the mixture *now*.

Equilibrium constant, K. At equilibrium the ratio equals K. Its value is fixed at a given temperature for the written reaction and chosen standard states.

If $Q < K$	If $Q = K$	If $Q > K$
The forward reaction forms A.	The reaction is at equilibrium.	The reverse reaction forms A_2.

Does doubling the pressure always double Q?

A question from Week 5. How does the pressure argument change for a real gas?

Start with the ideal-gas reaction $\mathrm{A_2(g)} \rightleftharpoons 2\mathrm{A(g)}$. At 350 K, $K = 3.953$ for $p^\circ = 100$ kPa. Compress the mixture from 100 to 200 kPa at fixed temperature. Consider the instant before the composition responds.

Starting value	Predict Q immediately after compression	Which way will the reaction move?
$Q = K = 3.953$	Calculate on paper	Toward A or A2?
$Q = 1.000$	Calculate on paper	Toward A or A2?

Take one minute. Compare your two answers with a neighbor.

Compare the new quotient with K

At fixed mole fractions, the ideal-gas quotient for this reaction is proportional to pressure.

$$Q = \frac{y_{\mathrm{A}}^2}{y_{\mathrm{A_2}}} \frac{P}{p^\circ}$$

Before compression	Immediately after	Response at 350 K
$Q = 3.953$	$Q = 7.906 > K$	More A2 forms.
$Q = 1.000$	$Q = 2.000 < K$	More A forms.

As the reaction proceeds, the mole fractions change. At the new equilibrium, $Q = K = 3.953$ in both cases.

Does gas non-ideality change K?

At the same temperature, for the same written reaction and standard states, **the thermodynamic K stays the same**.

$$K = \exp\left(-\frac{\Delta_r G^\circ}{RT}\right)$$

Fugacity coefficients enter the current mixture's activity ratio. For $\mathrm{A_2(g)} \rightleftharpoons \mathrm{2A(g)}$,

$$Q = \frac{\phi_{\mathrm{A}}^2}{\phi_{\mathrm{A_2}}} \frac{y_{\mathrm{A}}^2}{y_{\mathrm{A_2}}} \frac{P}{p^\circ}$$

At equilibrium, this **full ratio** equals K. Non-ideality changes the equilibrium composition needed to satisfy that condition. We will now calculate the fugacities that enter it.

Fugacity expresses the chemical potential

For an ideal gas mixture, the partial pressure $y_i P$ enters the chemical potential.

$$\mu_i^{\text{ideal}} = \mu_i^\circ + RT \ln\left(\frac{y_i P}{p^\circ}\right)$$

For a real gas, define the **component fugacity**, f_i, through the same logarithmic form and standard state.

$$\mu_i = \mu_i^\circ + RT \ln\left(\frac{f_i}{p^\circ}\right)$$

Fugacity has pressure units. The activity $a_i = f_i / p^\circ$ is dimensionless. In the ideal-gas limit, $f_i = y_i P$.

Definition reference · IUPAC

Compare fugacity with partial pressure

The **fugacity coefficient**, ϕ_i, is the ratio of component fugacity to its partial pressure.

$$\phi_i = \frac{f_i}{y_i P}, \qquad f_i = \phi_i y_i P$$

Subtract the ideal-mixture chemical potential at the **same** $T, P, \mathbf{y}$.

$$\mu_i - \mu_i^{\text{ideal}} = RT \ln \phi_i$$

At $\phi_i = 1$, $f_i = y_i P$ and the chemical potential matches the ideal-mixture value. A value below one gives a negative departure in chemical potential; a value above one gives a positive departure.

Each component has its own $\phi_i(T, P, \mathbf{y})$.

Calculate phi from an equation of state

An EOS supplies the partial-molar properties used to calculate $RT \ln \phi_i$. Choose the EOS and its parameters for the stated T, P, $\mathbf{y}$.

For the **low-density, pressure-form second-virial EOS**, used in Lab 09,

$$Z = \frac{PV}{nRT} = 1 + \frac{B_{\text{mix}} P}{RT}, \qquad B_{\text{mix}} = \sum_i \sum_j y_i y_j B_{ij}$$

The component fugacity coefficients follow from this mixture EOS.

$$\ln \phi_i = \left(2 \sum_j y_j B_{ij} - B_{\text{mix}} \right) \frac{P}{RT}$$

The coefficients $B_{ij}(T)$ come from gas-property data or a correlation. The cross coefficient describes unlike pairs. Use consistent units and check the EOS validity range.

Work out both coefficients from the EOS

Use synthetic inputs at 350 K and 200 kPa. Let component 1 be A and component 2 be A2, with $y_1 = 0.80$, $y_2 = 0.20$.

Take $B_{11} = -1.00$, $B_{12} = B_{21} = -0.600$, $B_{22} = -0.200$ L mol^{-1} and $R = 8.31446$ L kPa mol^{-1} K^{-1}.

$$B_{\text{mix}} = 0.80^2(-1.00) + 2(0.80)(0.20)(-0.600) + 0.20^2(-0.200) = -0.840 \text{ L mol}^{-1}$$

Component	$\sum_j y_j B_{ij}$ (L mol^{-1})	$\ln \phi_i$	$\phi_i = \exp(\ln \phi_i)$
A	-0.920	$(-1.000)(200)/(8.31446 \times 350) = -0.068727$	0.93358
A2	-0.520	$(-0.200)(200)/(8.31446 \times 350) = -0.013745$	0.98635

Carry these coefficients into the activity calculation. Keep unrounded values until the final result.

A real gas uses fugacity in each activity

At a specified T, P, $\mathbf{y}$, a mixture equation of state gives each component's fugacity coefficient ϕ_i. With an ideal-gas standard state,

$$a_i = \frac{f_i}{p^\circ} = \frac{\phi_i y_i P}{p^\circ}, \qquad Q = \frac{a_{\mathrm{A}}^2}{a_{\mathrm{A}_2}}.$$

For this dissociation, the multiplier on the ideal-gas estimate is $\phi_{\mathrm{A}}^2 / \phi_{\mathrm{A}_2}$. The coefficients can change during compression and as the reaction proceeds. The same T, written reaction and standard states give the same K.

Try it. Continue with the same EOS calculation at 350 K and 200 kPa, with $y_{\mathrm{A}} = 0.80$, $y_{\mathrm{A}_2} = 0.20$. Use the calculated coefficients $\phi_{\mathrm{A}} = 0.93358$ and $\phi_{\mathrm{A}_2} = 0.98635$, with $p^\circ = 100$ kPa. Calculate both fugacities, both activities and the reaction direction.

Work the real-gas example on paper

The pressure ratio is $200/100 = 2$. At the specified composition,

Quantity	Calculation	Value
f_{A}	$\phi_{\mathrm{A}}(0.80)(200)$	149.373 kPa
$f_{\mathrm{A_2}}$	$\phi_{\mathrm{A_2}}(0.20)(200)$	39.454 kPa
a_{A}	$f_{\mathrm{A}}/(100\ \mathrm{kPa})$	1.49373
$a_{\mathrm{A_2}}$	$f_{\mathrm{A_2}}/(100\ \mathrm{kPa})$	0.394539
Q	$a_{\mathrm{A}}^2/a_{\mathrm{A_2}}$	5.6553

The ideal-gas estimate at this same state is 6.40. Both exceed $K = 3.953$, so A2 forms. To find the final extent for the real gas, update the composition and both fugacity coefficients until $Q = K$. Lab 12 solves the ideal-gas case.

The 180-minute class

Class time	Activity
0–20 min	Derive Q and K, then calculate fugacity and check the pressure question
20–40 min	Revisit Weeks 2–5 in pairs with paper and a calculator
40–65 min	Balance a cell reaction and calculate its potential by hand
65–75 min	Demonstrate the concentration cell in Lab 13
75–85 min	Break
85–155 min	Pair investigation and independent check
155–180 min	Discuss results (15), exit ticket (5), feedback (3), closing (2)

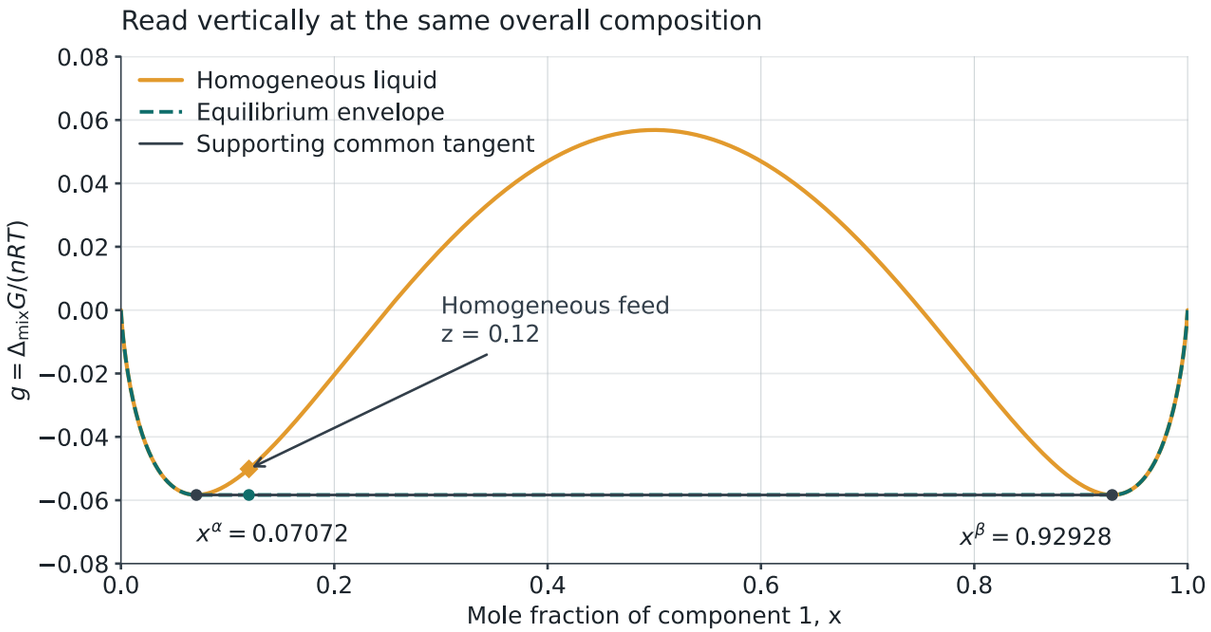
Week 2 · What did the VLE measurement tell us?

A VLE measurement gives $P = 100$ kPa, $x_1 = 0.40$, $y_1 = 0.60$. Assume an ideal vapor, negligible Poynting correction and a pure-liquid reference state.

$$y_1 P = x_1 \gamma_1 P_1^{\mathrm{sat}}, \qquad a_1 = x_1 \gamma_1 .$$

1. Use $P_1^{\mathrm{sat}} = 120$ kPa. Find γ_1 and a_1.
2. A second source gives 150 kPa at the same temperature. Predict which way your inferred γ_1 moves.
3. Before fitting the data, which Antoine conventions would you check?

Week 3 · Would this feed split into two liquids?



Mixing Gibbs energy and the supporting common tangent for the Week 3 Margules liquid at A=3.

At $z_1 = 0.12$, compare the curve and the supporting line at the **same overall composition**. Which state has lower Gibbs energy?

The homogeneous curve has positive local curvature here. What does the tangent tell you that curvature alone cannot?

A Gibbs–Duhem area test examines the VLE data. How could errors of opposite sign affect its signed integral?

Week 4 · Can the vessel supply that much gas?

At 300 K, a 1.00 L vessel contains 0.00200 mol of volatile A. Its equilibrium sublimation pressure is 10.0 kPa. Use $R = 8.314$ kPa L mol^{-1} K^{-1} and assume an ideal gas.

1. Calculate the gas inventory required for $P_A = 10.0$ kPa. Will any bulk solid remain?
2. Calculate the actual P_A. What happens to P_A if you add an inert gas at fixed T, V?
3. In a separate adsorption vessel, an adsorbent of mass m_s holds $q(P_A, T)$ mol per kg. Which term must you add to the gas inventory to close the balance?

Week 5 · Change pressure, then change temperature

Lab 12 used 1 mol of A₂ initially, zero A, no inert, and $p^\circ = 100$ kPa. Use the same ideal-gas reaction in all three cases.

Case	T (K)	P (kPa)	K	Equilibrium extent (mol)
12A	350	100	3.953	0.7050
12B	350	400	3.953	0.4451
12C	400	100	33.861	0.9457

Which pair isolates pressure? Which isolates temperature? For 12C assume a constant positive reaction enthalpy of 50 kJ mol^{-1}. Explain the change in K.

With a partner, use the extent in 12B to check $2n_{\mathrm{A_2}} + n_{\mathrm{A}}$ against the feed.

Check your four predictions

Week	Compare with your calculation or prediction
2 · VLE	With $P_1^{\mathrm{sat}} = 120$ kPa, $\gamma_1 = 1.25$ and $a_1 = 0.50$. With $P_1^{\mathrm{sat}} = 150$ kPa they become 1.00 and 0.40. Check Antoine units, log base and temperature range.
3 · Stability	At $z_1 = 0.12$ the two-liquid line lies lower. Positive local curvature describes small fluctuations. Opposite errors can cancel in the signed area test.
4 · Inventory	Saturation requires 0.00401 mol. All 0.00200 mol enters the gas, giving $P_A = 4.99$ kPa. Adding ideal inert at fixed T, V raises total pressure while P_A stays at 4.99 kPa. For adsorption add $m_s q$ mol.
5 · Reaction	12A–12B isolate pressure. 12A–12C isolate temperature. Heating raises K for this endothermic reaction. In 12B, $2(0.5549) + 0.8902 = 2.0000$ mol of A atoms.

The question for today

What determines the sign of a concentration-cell potential?

Week 5 gave us $\Delta_r G = RT\ln(Q/K)$. A reversible cell turns that reaction driving force into a potential difference, $E = -\Delta_r G/(nF)$.

Predict on paper. For the reaction as written, what sign should E have when $Q < K$? What voltage would you measure at complete cell equilibrium, $Q = K$?

Pre-class reading and preparation questions

Two reduction half-reactions

Enter both half-reactions as reductions. Reverse the left reaction and add the right after matching electron counts.

$$E^\circ_{cell} = E^\circ_{right} - E^\circ_{left}$$

Both potentials need the same reference electrode and temperature. Multiplying a half-reaction does not multiply its potential.

Nernst equation from reaction thermodynamics

Combine ΔrG=ΔrG°+RTlnQ with ΔrG=−nFE:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298.15 K,

$$E = E^\circ - \frac{0.05915935\ \mathrm{V}}{n} \log_{10} Q$$

Changing T requires valid E° data at the new T.

A concentration-cell calculation

Use the same $\mathrm{M}^{2+}/\mathrm{M}$ couple on both sides at 298.15 K. For oxidation on the left and reduction on the right, $Q = a_{\mathrm{L}}/a_{\mathrm{R}}$. Take $a_{\mathrm{L}} = 0.01$, $a_{\mathrm{R}} = 1$, $n = 2$, and neglect the junction potential.

$$Q = 0.01, \qquad E = +0.05915935\ \mathrm{V}$$

E°=0, K=1, but E is nonzero. Reversing the concentrations changes the sign of E. Equal activities give E=0.

ΔrG≈−11.416 kJ per mol of the written reaction.

An independent check

Use dimensionless activities. Concentration is an approximation with a stated standard concentration and activity coefficient.

Use Eright−Eleft and deltaG=−nFE with the balanced electron count. Equal activities must give zero potential for the same reduction couple.

One demonstration · 10 minutes

Open the core lab

Open the concentration-cell preset on the projector. On paper, make three rows for the original, exchanged and equal activities. If your device is offline, use the projected inputs and complete the same hand check. Predict the sign, exchange left and right activities, then set them equal. Keep T and reference conventions fixed.

Keep one saved baseline for the pair investigation.

Pair investigation · 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E, then independently check deltaG=−nFE.

Discussion · 15 minutes

What determines the sign of a concentration-cell potential?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

After class · 30–40 minutes

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E, then independently check deltaG=−nFE.

Assessment 6 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path · Offline package

Exit ticket + feedback · 8 minutes

5 min · Explain and check

Answer the session question in the form. Give one result or independent check from today’s lab, with units.

3 min · Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for **“Response received.”** We will use the final 2 minutes to close the class.

Scan to submit Session 6 exit ticket and feedback.

Open Session 6 form